Chapter 46: Similarity of linear operators Worksheet

1. Are each of the following statements true or false?

(a) If $M_1 = M_B(T)$ and $M_2 = M_D(T)$ are two matrix representations for the same linear operator $T : V \to V$, then M_1 and M_2 have the same trace.

(b) If $T : V \to V$ by $T(\mathbf{v}) = c\mathbf{v}$ for some scalar c, then T is diagonalizable.

(c) If $T : P_1 \to P_1$ has an eigenvalue of 0, then T is not an isomorphism.

(d) If $T : V \to V$ is an isomorphism, then T is diagonalizable.

2. Define $T : P_1 \to P_1$ as the linear transformation $T\,(ax + b) = (b - 2a)x + (3b + a)$. Let $B = (1, x)$, and $D = (x, x + 1)$.

(a) Find the matrix of T corresponding to B, i.e. $M_B(T)$.

(b) Find the matrix of T corresponding to D, i.e. $M_D(T)$.

(c) Find the change matrix $P = P_{B \leftarrow D}$.

(d) Verify that $M_D(T) = P^{-1} M_B(T) P$.

(e) Use your work to justify: Is T an isomorphsim?

3. Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be the linear transformation whose matrix with respect to the standard basis B for $\mathbb{R}^3$ is $M_B(T) = A = [a_{ij}]$. Let $P = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$, and $A' = P^{-1}AP$. Find a basis D for $\mathbb{R}^3$ so that $A' = M_D(T)$.

4. Consider $T : P_1 \to P_1$ by $T(p(x)) = 2p(0)x + p'(x)x + p'(x)$.

(a) By choosing a basis B for P_1 and building $M_B(T)$, find the eigenvalues of T and a basis for each eigenspace. Note that your basis elements should be "translated" back to P_1.

(b) Is T diagonalizable? If so, give a basis D for P_1 where $M_D(T)$ is a diagonal matrix.

5. Consider the matrix transformation $L : \mathbb{R}^3 \to \mathbb{R}^3$, $L(\mathbf{x}) = A\mathbf{x}$ with respect to standard basis S, where the matrix A is

$$A = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 3 & 1 \\ 0 & 0 & 2 \end{bmatrix}.$$

(a) Find the eigenvalues and corresponding eigenvectors of A.

(b) In Part 1, there should be three different eigenvalues. For each eigenvalue, pick one of its eigenvectors and show that the resulting set is linearly independent.

(c) Suppose $\mathbf{v}_1^{'}$, $\mathbf{v}_2{'}$ and $\mathbf{v}_3{'}$ are eigenvectors chosen in Part 2 and let $S^{'} = \{\mathbf{v}_1^{'}, \mathbf{v}_2^{'}, \mathbf{v}_3^{'}\}$ be an ordered basis of $\mathbb{R}^3$ (please recall why we can assert they form a basis of $\mathbb{R}^3$), find the representation of L with respect to $S^{'}$, called $M_{S'}(L)$.